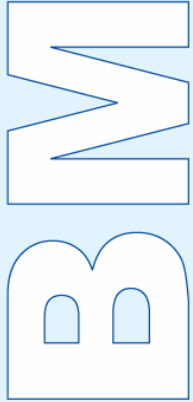




MiDAS Family



IP Specification of MiDAS1.0: Turbo 80C52

V1.4

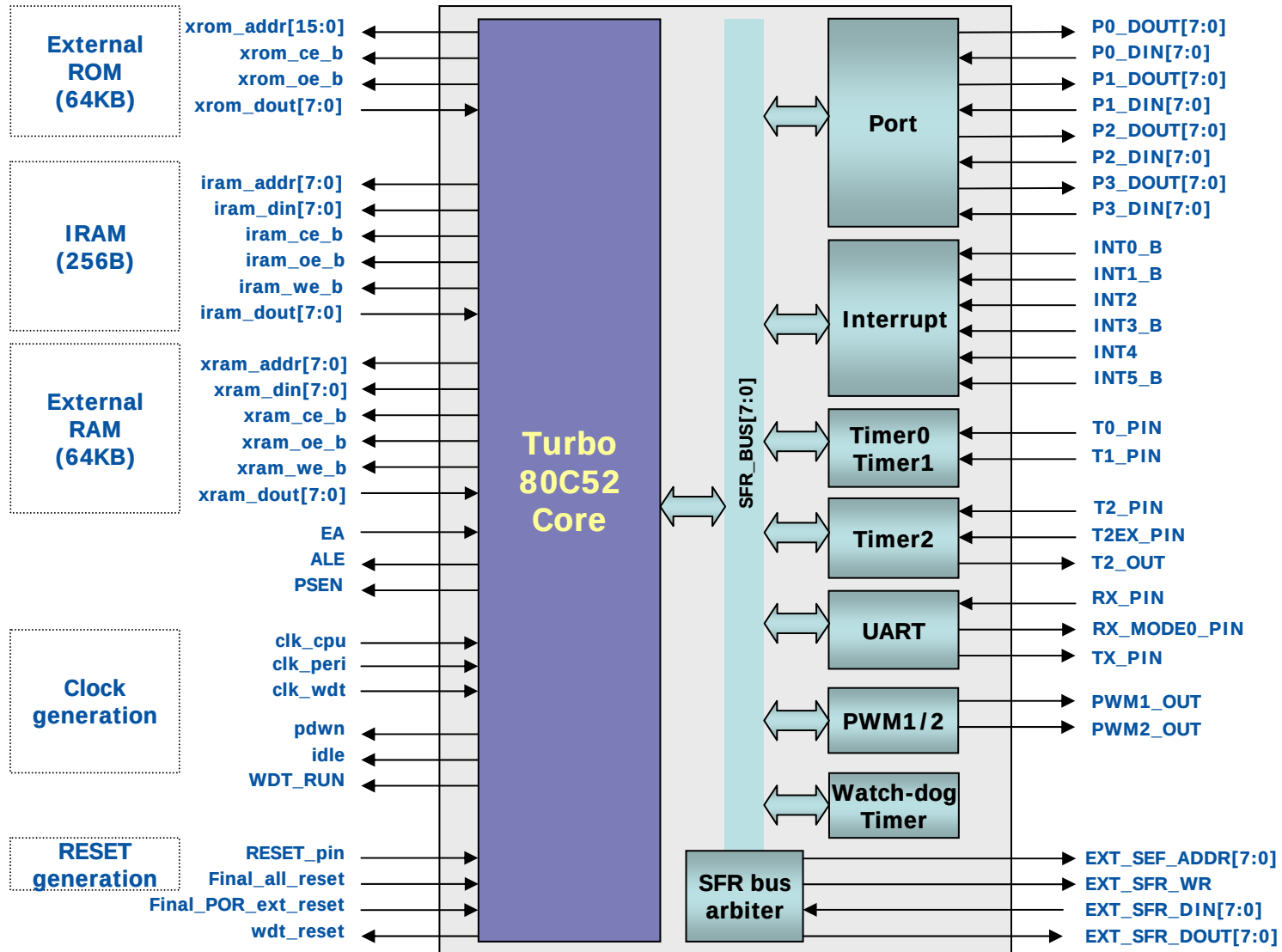
June 2007

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MiDAS1.0 IP features

- ◆ CPU
 - ✓ 8-bit turbo 80C52 architecture
 - ✓ 4 cycles/1 machine cycle
 - ✓ Instruction level compatible with Intel 80C52
- ◆ Max 64Kbytes Program Memory interface
- ◆ Max 64Kbytes data memory interface
- ◆ Operating Frequency
 - ✓ Max. 50MHz @0.5 um process
 - ✓ Max. 100MHz @0.18 um process
- ◆ Fully programmable 32 IO pins
 - ✓ Optional Input/Output control
- ◆ Three 16-bit Timer/Counters
- ◆ 2 channel 8-bit programmable PWM
- ◆ 27bit programmable watch-dog timer (WDT)
- ◆ Full-Duplex UART
 - ✓ Automatic address recognition
- ◆ 11 interrupt sources (including 6 external sources)
 - ✓ Timer0/1/2, UART, WDT, 6 External
 - ✓ Four-level interrupt priority
- ◆ Wake-up from power-down mode by RESET or external interrupt 0/1 and watch-dog timer interrupt

Block diagram



SFR mapping

F8h	EIP								FFh
F0h	B								F7h
E8h	EIE								EFh
E0h	ACC								E7h
D8h	WDON				PWM1CON	PWM2CON	PWM1D	PWM2D	DFh
D0h	PSW								D7h
C8h	T2CON	T2MOD	RCAP2L	RCAP2H	TL2	TH2			CFh
C0h					PMR				C7h
B8h	IP	SADEN							BFh
B0h	P3							IPH	B7h
A8h	IE	SADDR							AFh
A0h	P2								A7h
98h	SCON	SBUF							9Fh
90h	P1	EXIF							97h
88h	TCON	TMOD	TL0	TL1	TH0	TH1	CKCON		8Fh
80h	P0	SP	DPL	DPH				PCON	87h

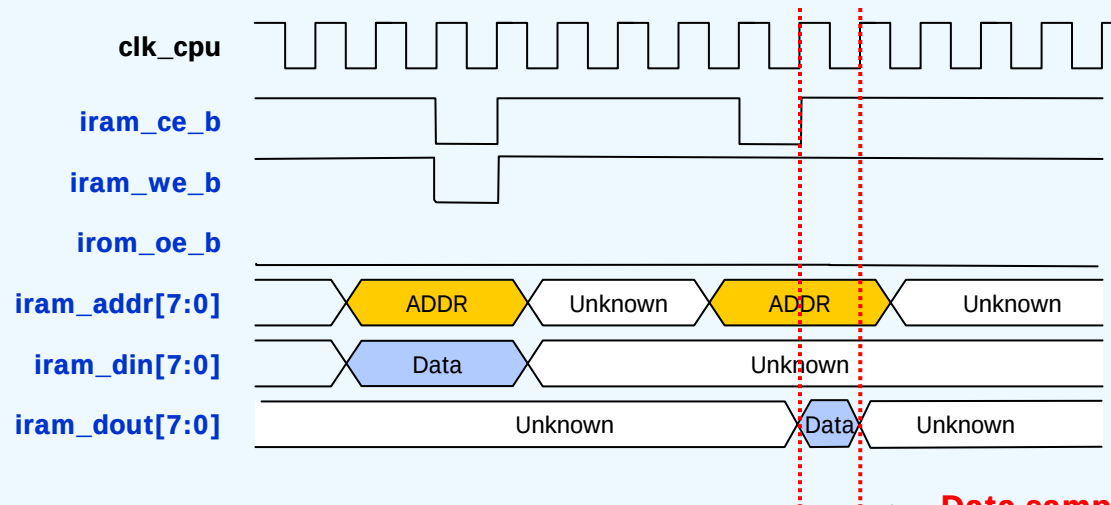


: Bit Addressable SFR

Internal RAM(256Bytes) interface

Signal name	Direction	Description
iram_addr[7:0]	Output	Internal RAM address
iram_din[7:0]	Output	Internal RAM input data
iram_ce_b	Output	Internal RAM enable. Low active
iram_we_b	Output	Internal RAM write enable. Low active
iram_oe_b	Output	Internal RAM output enable. Low active
iram_dout[7:0]	input	Internal RAM output data

Internal RAM Write & Read Timing

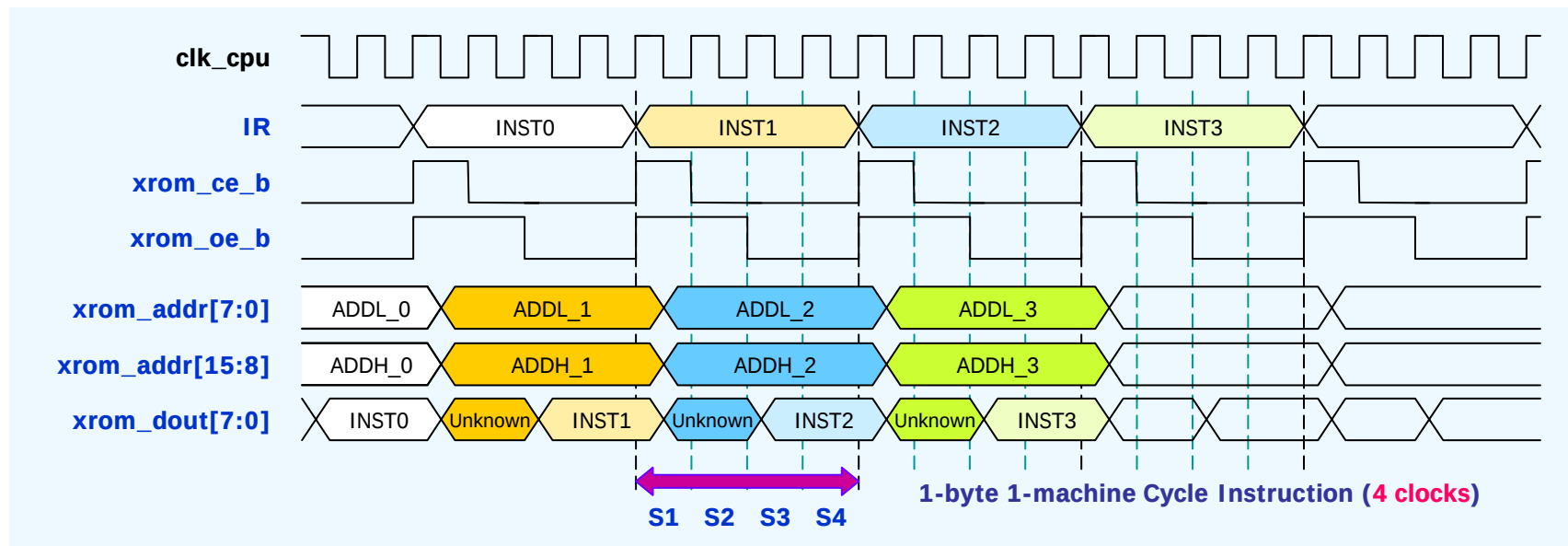


Data sampling

(iram_dout is always enabled, this timing is for synchronous RAM)

External ROM(64Kbytes) interface

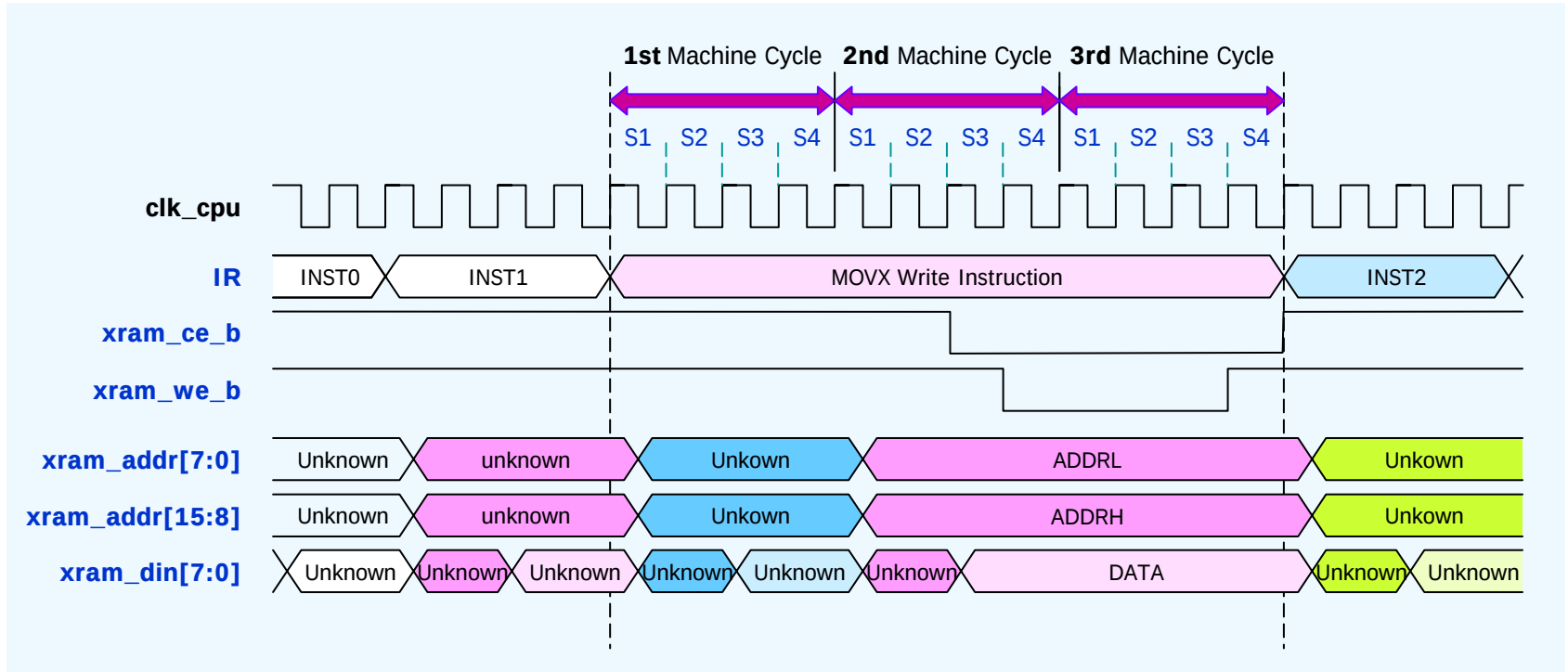
Signal name	Direction	Description
xrom_addr[15:0]	Output	External ROM address
xrom_ce_b	Output	External ROM enable. Low active
xrom_oe_b	Output	External ROM write enable. Low active
xrom_dout[7:0]	Input	External ROM output data



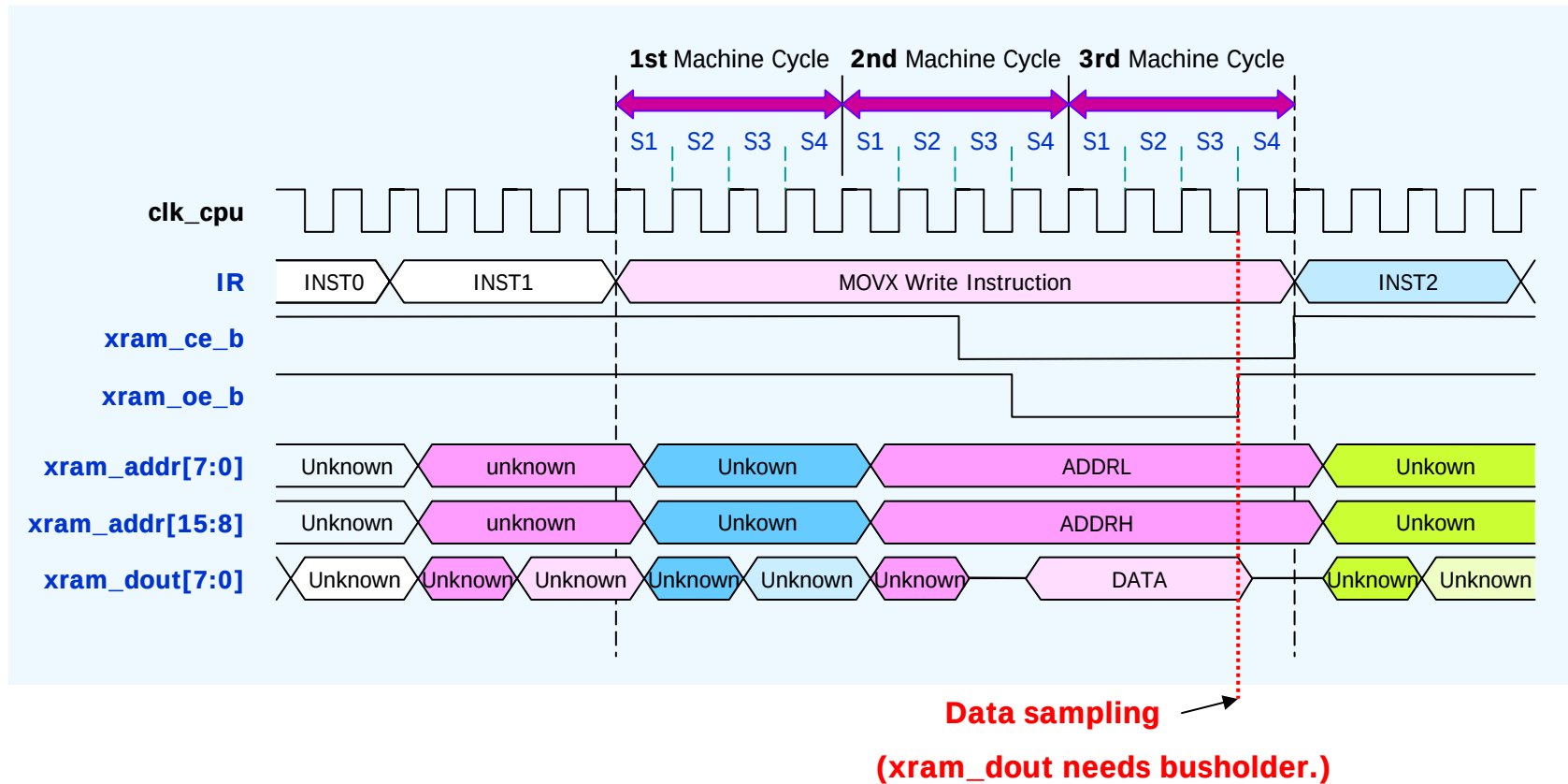
External RAM(64KBytes) interface

Signal name	Direction	Description
xram_addr[15:0]	Output	External RAM address
xram_din[7:0]	Output	External RAM input data
xram_ce_b	Output	External RAM enable. Low active.
xram_we_b	Output	External RAM write enable. Low active
xram_oe_b	Output	External RAM output enable. Low active
xram_dout[7:0]	input	External RAM output data

External RAM Write Timing (MOVX)



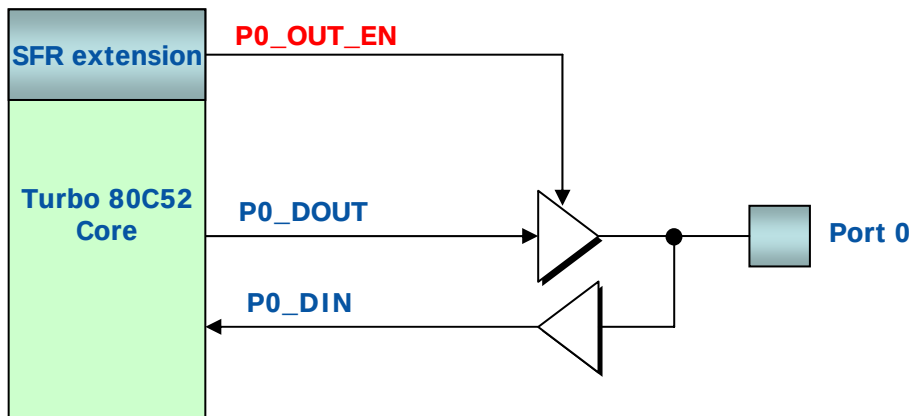
External RAM Read Timing (MOVX)



Port0/1/2/3 interface

Signal name	Direction	Description
P0_DOUT[7:0]	Output	Port 0 output data (SFR of P0)
P0_DIN[7:0]	Input	Port 0 input data
P1_DOUT[7:0]	Output	Port 1 output data (SFR of P1)
P1_DIN[7:0]	Input	Port 1 input data
P2_DOUT[7:0]	Output	Port 2 output data (SFR of P2)
P2_DIN[7:0]	Input	Port 2 input data
P3_DOUT[7:0]	Output	Port 3 output data (SFR of P3)
P3_DIN[7:0]	Input	Port 3 input data

Example : Bi-directional port implementation



//Example of Coding

//P0DIR is extended SFR.

//P0DIR is allocated to 0xF9 in SFR.

P0DIR EQU 0F9h

//When P0DIR[x] = "0", I want to set output mode.

MOV P0DIR, #00h

MOV P0, #55h //55h will drive to Port 0.

//Now I want to set input mode.

MOV P0DIR, #0FFh

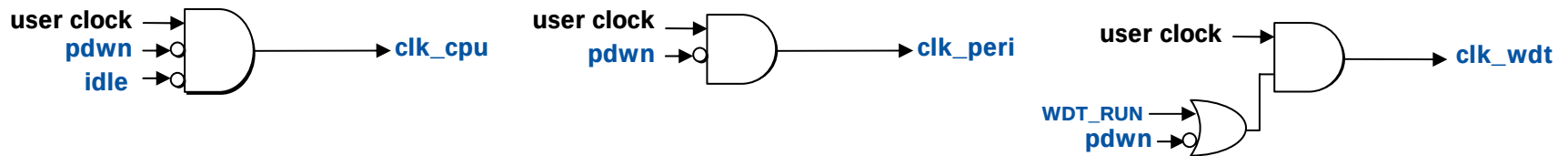
MOV A, P0 //ACC <= Port 0

Reset & Clock interface

Signal name	Direction	Description
clk_cpu	Input	Clock for CPU block
clk_peri	Input	Clock for Peripheral block
clk_wdt	Input	Watch-dog timer clock.
WDT_RUN	Output	Watch-dog timer is now running. So, WDT is alive even if PDWN = 1. (power-down mode)
pdwn	Output	When power-down mode, pdwn will be "1".
idle	Output	When idle mode, idle will be "1".

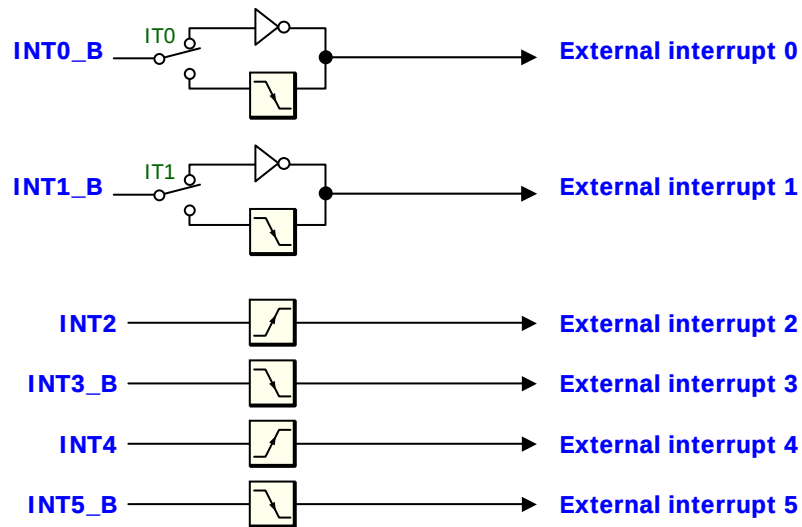
- All MiDAS1.0 IP design is synchronized at posedge clock.
- clk_cpu and clk_peri are on the same phase.
- clk_wdt_b is used only for watch-dog timer.

Example : clock generation



Interrupt interface

Signal name	Direction	Description
INT0_B	Input	External interrupt 0. Low active. Low level or negedge detection.
INT1_B	Input	External interrupt 1. Low active. Low level or negedge detection.
INT2	Input	External interrupt 2. High active. Only posedge detection.
INT3_B	Input	External interrupt 3. Low active. Only negedge detection.
INT4	Input	External interrupt 4. High active. Only posedge detection.
INT5_B	Input	External interrupt 5. Low active. Only negedge detection.



[Interrupt Vector Address]

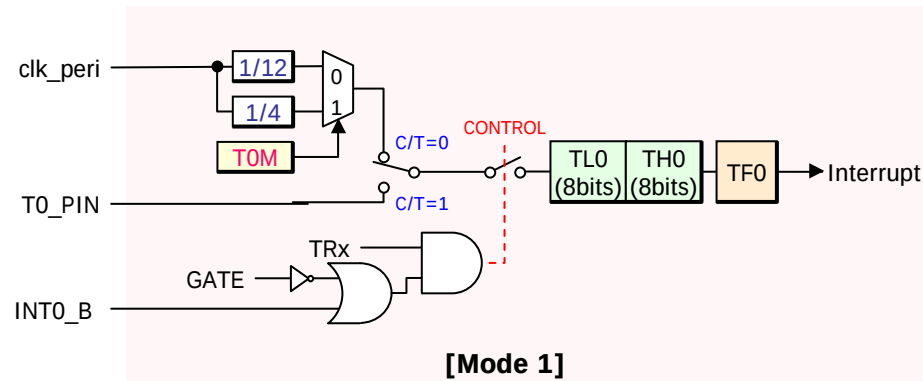
HIGH
↑
PRIORITY
↓
LOW

Interrupt Sources	Address	Priority Level
INT0_B	0003h	4 Levels
TF0	000Bh	4 Levels
INT1_B	0013h	4 Levels
TF1	001Bh	4 Levels
RI + TI	0023h	4 Levels
TF2	002Bh	4 Levels
INT2	0043h	2 Levels
INT3_B	004Bh	2 Levels
INT4	0053h	2 Levels
INT5_B	005Bh	2 Levels
WDT	0063h	2 Levels

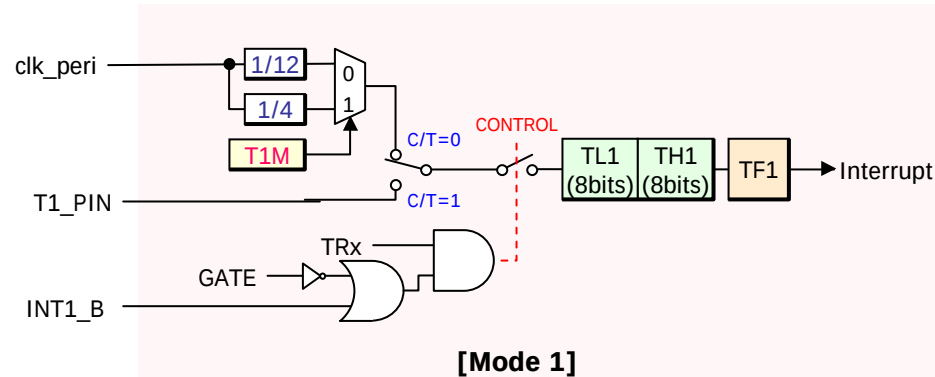
Timer 0/1 interface

Signal name	Direction	Description
T0_PIN	Input	Timer 0 input clock
T1_PIN	Input	Timer 1 input clock

Example 0 : Timer 0 connection



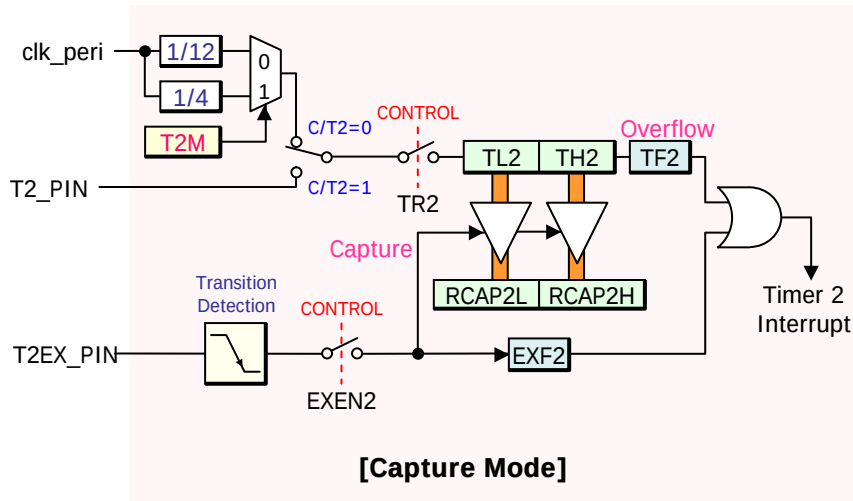
Example 1 : Timer 1 connection



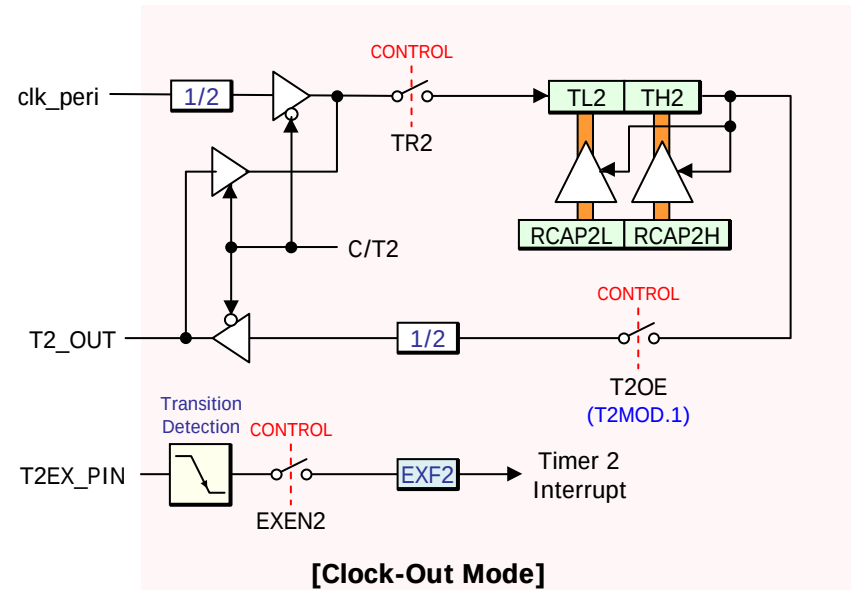
Timer 0/1 interface

Signal name	Direction	Description
T2_PIN	Input	Timer 2 input clock
T2EX_PIN	Input	Timer 2 capture enable and interrupt flag setting. Negedge active.
T2_OUT	Output	Timer 2 overflow output

Example 0 : T2_PIN/T2EX_PIN connection



Example 1 : T2_OUT connection

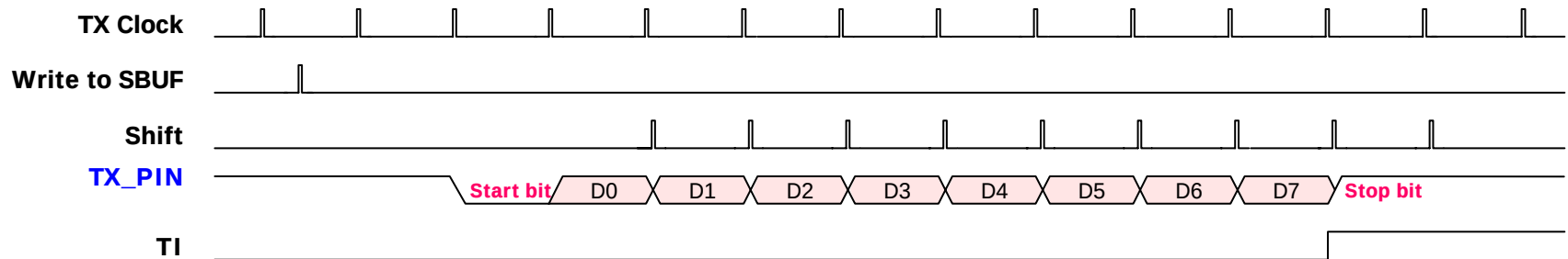


UART interface

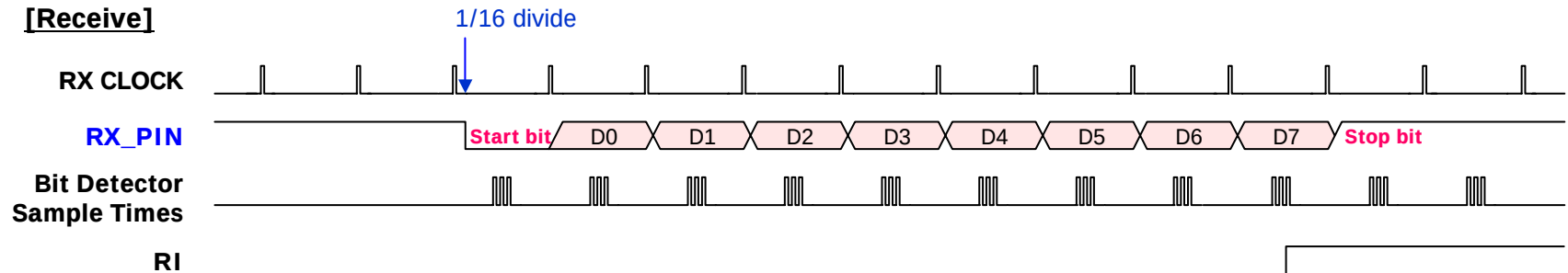
Signal name	Direction	Description
TX_PIN	Output	UART transmit data
RX_PIN	Input	UART receive data
RX_MODE0_PIN	Output	UART RXD output when operates in MODE0

Example : TX_PIN and RX_PIN function in UART mode 1.

[Transmit]



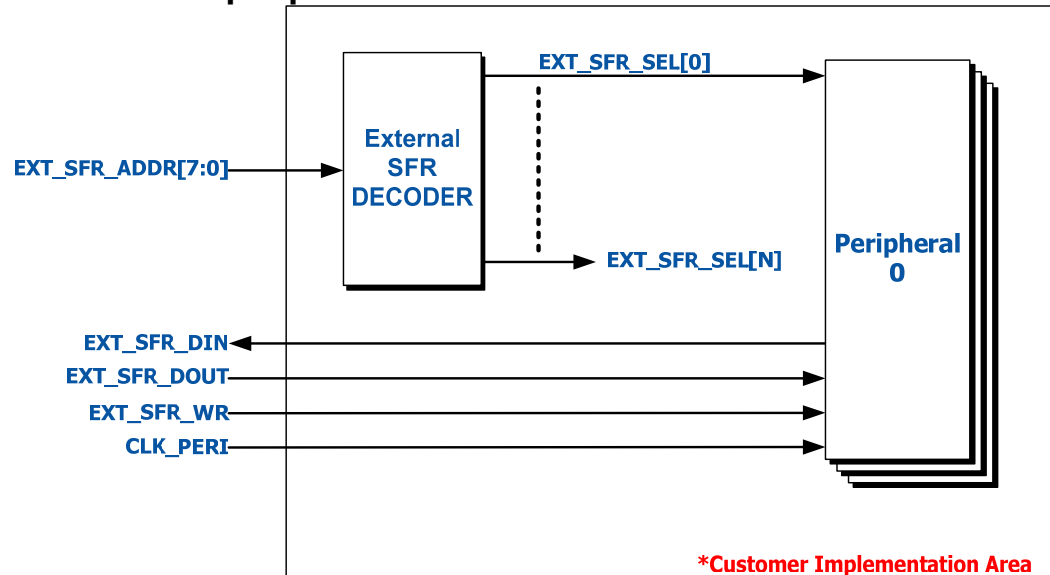
[Receive]



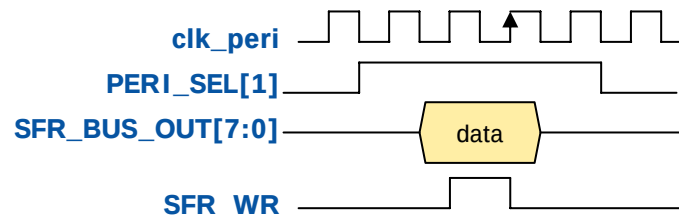
Additional Peripheral interface

Signal name	Direction	Description
EXT_SFR_DOUT[7:0]	Output	SFR bus output data
EXT_SFR_DIN[7:0]	Input	SFR data input data (When do not used, must be tied to 8'h00)
EXT_SFR_ADDR[7:0]	Output	SFR address
EXT_SFR_WR	Output	SFR write enable. Active High.

Example 0: Additional peripheral connection



Example 1: Additional peripheral SFR write timing



Etc. interface

Signal name	Direction	Description
EA	Input	External ROM access enable. If EA=0, external ROM access. If EA=1, Internal ROM access.
ALE	Output	Address latch enable.
PSEN	Output	External Program memory strobe enable.